

Computational Safety for Generative AI: A Hypothesis Testing Perspective

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Abstract

AI safety is a rapidly growing area of research that seeks to prevent the harm and misuse of frontier AI technology, particularly with respect to generative AI (GenAI) tools that are capable of creating realistic and high-quality content through text prompts. Examples of such tools include large language models (LLMs) and text-to-image (T2I) diffusion models. As the performance of various leading GenAI models approaches saturation due to similar training data sources and neural network architecture designs, the development of reliable safety guardrails has become a key differentiator for responsibility and sustainability. This paper presents a formalization of the concept of *computational safety*, which is a mathematical framework that enables the quantitative assessment, formulation, and study of safety challenges in GenAI through the lens of signal processing theory and methods. In particular, we explore two exemplary categories of computational safety challenges in GenAI that can be formulated as hypothesis testing problems. For the *safety of model input*, we show how sensitivity analysis and loss landscape analysis can be used to detect malicious prompts with jailbreak attempts. For the *safety of model output*, we elucidate how statistical signal processing can be used to detect AI-generated content. Finally, we discuss key open research challenges, opportunities, and the essential role of signal processing in computational AI safety.

1. Introduction

Signal processing for pattern recognition and machine intelligence has played a pivotal role in ensuring the stability, security, and efficiency of numerous engineering systems and information technologies, including, but not limited to, telecommunications, information forensics and security,

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machine learning, data science, and control systems. With the recent advances, wide accessibility, and deep integration of generative AI (GenAI) tools into our society and technology, such as ChatGPT and the emerging agentic AI applications, understanding and mitigating the associated risks of the so-called “frontier AI technology” is essential to ensure a responsible and sustainable use of GenAI. In addition, as the performance of state-of-the-art GenAI models surpasses that of an average human in certain tasks, but reaches a plateau in standardized capability evaluation benchmarks due to similar training data sources and neural network architecture designs (e.g., the use of decoder-only transformers), improving and ensuring safety is becoming the new arms race among GenAI stakeholders. In addition to ongoing activities in risk governance, policy-making, and legal regulation (e.g., EU AI Act, AI safety/security institutes, etc.), there are growing concerns about the broader socio-technical impacts (Qi et al., 2024). Notable examples include breaking the embedded safety alignment of GenAI models to induce misbehavior (e.g., asking large language models how to perform illegal or dangerous activities) (Zou et al., 2023) or generating inappropriate content, and misusing GenAI tools to create and spread misinformation (e.g., using text prompts to create deepfakes and influence public opinion) (Verdoliva, 2020).

The main goal of this paper is to present the central message in a technically sound and rigorous manner as follows: *While AI safety can be seen as a significant challenge in the context of modern GenAI technology, it is possible to reformulate many of the associated problems as a standard hypothesis testing task.* Furthermore, insights and methods developed in signal processing can be applied to improve AI safety. Specifically, we define *computational AI safety* as the set of safety problems that can be formulated as hypothesis-testing tasks in signal processing. We also provide examples of how signal processing techniques such as sensitivity analysis, subspace projection (Vaswani et al., 2018), loss landscape (Li et al., 2018), and adversarial learning can be used to study and improve AI safety.

Figure 1 provides an overview of our proposed signal processing framework for computational AI safety. Without loss of generality, we divide a GenAI system into three parts: the *model input* that processes a user’s query, the *GenAI model* that performs inference and action (for agentic

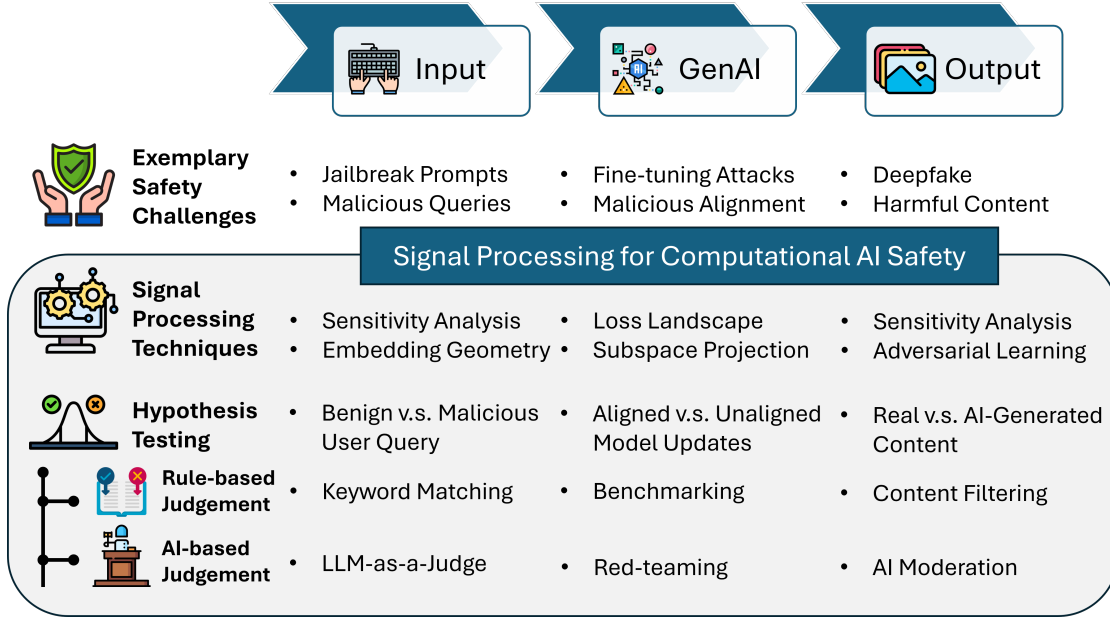


Figure 1. Overview of the framework on signal processing for computational AI safety. The two safety challenges highlighted for model input and output safety in GenAI are the detection of unsafe queries and AI-generated content. We define computational AI safety as the set of safety problems that can be formulated as hypothesis testing tasks in signal processing. We also provide examples of how signal processing techniques such as sensitivity analysis, and subspace projection can be used to improve AI safety. What is unique about GenAI is that the validity of the safety hypothesis (e.g., whether the input or output is safe) requires an additional judge function for certification, which can be either a rule-based approach (e.g., keyword matching) or an AI-based evaluation (e.g., LLM-as-a-judge or external contextual classifiers). The framework can be expanded with additional signal processing methods and system-level designs.

workflows), and the *model output* produced by the model given the input. The two representative safety challenges highlighted for model input and output safety in GenAI are the detection of unsafe queries and AI-generated content. It is worth noting that what is unique about GenAI is that the validity of the safety hypothesis (e.g., whether the input or output is safe) requires an additional judge function for certification, which can be either a rule-based approach (e.g., keyword matching) or an AI-based evaluation (e.g., LLM-as-a-judge or external contextual classifiers). For example, when assessing the risk of jailbreak attacks, one can use refusal-related keywords to quantify the rejection rate in GenAI, or use another LLM as a judge to provide a numerical rating of the safety level of the response.

To bridge AI safety and signal processing, this paper presents an overview of computational safety for GenAI, consolidates numerous computational AI safety problems through the lens of a unified hypothesis testing framework in signal processing, and provides comprehensive technical information to elucidate how signal processing methodologies can facilitate and enhance the development of AI safety tools. This paper’s signal-processing-centric perspective differs from the broad range of recent progress in attacks and defenses for GenAI in existing AI safety literature, such as (Chua et al., 2024; Ma et al., 2025; Bereska & Gavves,

2024), or in-depth discussions of AI safety in particular domains (Sangwan et al., 2023; Ullrich et al., 2024). The rest of the paper is organized as follows. First, we provide background on mainstream GenAI models and preliminaries on hypothesis testing and common evaluation metrics. Next, we formally introduce the signal processing framework for computational AI safety. We then examine two representative use cases in AI safety – jailbreak detection and mitigation, and AI-generated content. Finally, we provide our perspective on the position of AI safety and discuss several ongoing research challenges and opportunities at the intersection of signal processing and AI safety.

2. Background and Preliminaries

2.1. Mainstream GenAI Models

This paper focuses on the safety challenges of two mainstream GenAI models: Large Language Models (LLMs) and Diffusion Models (DMs). LLMs are autoregressive neural networks built on self-attention-based transformers with billions of trainable parameters, which dominate text-to-text generation applications (e.g., ChatGPT) and underpin many multi-modal GenAI tools, such as vision-language models, audio-language models, and code-language models. DMs are state-of-the-art neural networks capable of high-quality

text-to-image generation through a mathematical diffusion process, with extensions to other modalities such as video, audio, and molecular dynamics. The deployment of LLMs and DMs has two modes: direct use of the off-the-shelf GenAI model, or customized fine-tuning with some task-specific downstream datasets. Throughout this paper, we will use θ to denote the set of trainable parameters (model weights) for GenAI models.

Large language models (LLMs) The development of LLMs has two primary phases: pre-training and alignment. LLM pre-training refers to unsupervised machine learning on a large dataset (e.g., text extracted from the Internet) denoted by $\mathcal{D}_p$. Let $\mathbf{x} = \{x_1, x_2, \dots, x_T\}$ denote a sequence of tokens (i.e., “words” defined in the LLM vocabulary) randomly drawn from $\mathcal{D}_p$. For pre-training, the Next-token prediction (NTP) loss is commonly used, which is defined as the expected loss of the negative log-likelihood

$$\text{Loss}_{\text{NTP}} = -\mathbb{E}_{\mathbf{x} \sim \mathcal{D}_p} \frac{1}{|\mathbf{x}|} \sum_{t=1}^{|\mathbf{x}|} \log p_{\theta}(x_t | \mathbf{x}_{<t}), \quad (1)$$

where $|\mathbf{x}|$ denotes the token length of $\mathbf{x}$, $\mathbf{x}_{<t} := \{x_1, x_2, \dots, x_{t-1}\}$ denotes the tokens preceding the t -th token and $\mathbf{x}_{<1} := \emptyset$ (an empty set), $p_{\theta}(x_t | \mathbf{x}_{<t})$ is parameterized by a softmax function over all tokens. This loss is also known as *causal language modeling*, by using previous tokens to predict the next token. The family of LLMs trained with NTP loss is also called autoregressive LLMs. After pre-training on θ , an LLM undergoes the alignment process to teach the model to follow instructions and value alignment. Safety alignment is also implemented in this stage to guide the model in refusing to respond to unsafe queries. Specifically, for supervised fine-tuning (SFT) (or instruction-tuning), let $(\mathbf{x}, \mathbf{y})$ denote a pair of model input and the desired model output randomly drawn from a dataset $\mathcal{D}_a$, the SFT loss is used to update the pre-trained model weights by generalizing the NTP loss as

$$\text{Loss}_{\text{SFT}} = -\mathbb{E}_{(\mathbf{x}, \mathbf{y}) \sim \mathcal{D}_a} \frac{1}{|\mathbf{y}|} \sum_{t=1}^{|\mathbf{y}|} \log p_{\theta}(y_t | \mathbf{x}, \mathbf{y}_{<t}). \quad (2)$$

Here, $\mathbf{x}$ can also be conceived as the *context*, which is used to model the conditional probability of generating $\mathbf{y}$ given $\mathbf{x}$ as $p_{\theta}(\mathbf{y} | \mathbf{x}) = \prod_{t=1}^{|\mathbf{y}|} p_{\theta}(y_t | \mathbf{x}, \mathbf{y}_{<t})$. In addition to SFT, other forms of alignment include strategies such as reinforcement learning with human feedback (RLHF) (Ouyang et al., 2022) and direct preference optimization (DPO) (Rafailov et al., 2024).

Diffusion Models (DMs) During training, DMs follow a mathematical diffusion process to gradually encode data samples into random Gaussian noises, and train a neural network to learn to decode (reconstruct) the data samples

based on multiple denoising steps. Furthermore, DMs can be integrated with text-based prompts to guide the generation process. We briefly explain the basics of DM as follows. Let t denote the diffusion step and let $\mathbf{x}_t$ be the data sample (or its latent representation) obtained via a forward diffusion process subject to noise injection. Given a text prompt (context) c , we denote $\epsilon_{\theta}(\mathbf{x}_t | c)$ as the noise generator parameterized by a neural network θ . A reverse diffusion process is also used to estimate the underlying noise. In its simplest form, the diffusion process is given by

$$\hat{\epsilon}_{\theta}(\mathbf{x}_t | c) = (1 - \lambda) \cdot \epsilon_{\theta}(\mathbf{x}_t | \emptyset) + \lambda \cdot \epsilon_{\theta}(\mathbf{x}_t | c), \quad (3)$$

where $\hat{\epsilon}_{\theta}(\mathbf{x}_t | c)$ signifies the ultimate noise estimation attained by utilizing the DM conditioned on c , $\lambda \in [0, 1]$ is a guidance weight, and $\epsilon_{\theta}(\mathbf{x}_t | \emptyset)$ represents the contribution of the unconditional DM. During inference, the DM initializes with a standard Gaussian noise $z_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ with zero mean and identity covariance matrix $\mathbf{I}$. The noise z_T is then denoised using $\hat{\epsilon}_{\theta}(\mathbf{x}_T | c)$ to obtain z_{T-1} . This procedure is repeated to generate the authentic data at $t = 0$. A commonly used training objective for DM is the mean-squared-error (MSE), defined as

$$\text{Loss}_{\text{MSE}} = \mathbb{E}_{t, \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})} [\|\epsilon - \epsilon_{\theta}(\mathbf{x}_t | c)\|_2^2]. \quad (4)$$

For simplicity, we omit the expectation over the training data $\{(\mathbf{x}, c)\}$ for DM training.

2.2. Hypothesis Testing and Common Evaluation Metrics

Generative Hypothesis Testing Hypothesis testing refers to statistical methods for data-driven validation among a set of mutually exclusive hypotheses, of which only one is true. It is a classic methodology that is widely used in signal processing, such as detection and estimation problems. In particular, binary hypothesis testing involves an alternative hypothesis $\mathcal{H}_1$ and a null hypothesis $\mathcal{H}_0$. In principle, a test statistic s is derived from the observed data samples and is compared to a threshold η for testing $\mathcal{H}_1$ versus $\mathcal{H}_0$. Unlike classic signal processing problems, what is unique about computational safety for GenAI is that a hypothesis of interest is context-dependent and usually cannot be explicitly defined in clear mathematical forms. For example, does the user query (model input) violate the usage policy, or does the model output contain any harmful content? To address this issue, it is common practice to employ a judge function $J(\mathbf{x}, \mathbf{y}) \in \{0, 1\}$ for hypothesis validation, which takes a pair of model input and output $(\mathbf{x}, \mathbf{y})$ and returns a binary label that serves as a proxy for the ground-truth hypothesis. The judge function J can be as simple as a rule-based system, such as flagging problematic context or content using keyword filtering. It can also be a complex and opaque decision-making process performed by another AI model, such as LLM-as-a-judge. However, these judge functions